WORKED SOLUTION

STEP I 1998, Question 11

Mechanics • Connected Particles Over Pulley

STEP I 1998, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q11

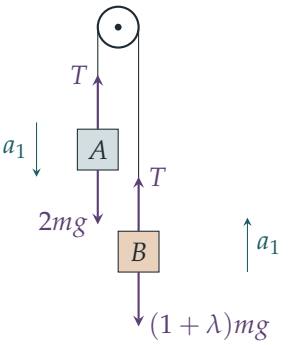
The key idea

Each round trip has two stages, and in each stage the masses are constant, so the acceleration is constant and we can use $s = \frac{1}{2}at^2$. On the way down, the full bucket of water outweighs the ore and bucket, and the pair moves as one Atwood machine. On the way back up, bucket A is empty and massless, so nothing holds B back: it simply falls freely. For the second part, "most ore in a very long time" means "largest ore per cycle divided by time per cycle".

Solution

The time for one cycle

Model: the buckets and their contents are particles, the chain is light and inextensible, and the pulley is small, smooth and light. So the tension T is the same at both ends of the chain, and both buckets move with the same speed. When A is at the top, B is at the bottom (the chain has length l), so each stage moves each bucket a distance l.



Stage 1: A descends. Bucket A carries $2m$ of water; bucket B has total mass $m + \lambda m = (1 + \lambda)m$. Let a_1 be the common acceleration (A down, B up). Newton's second law for

each bucket gives

$$\begin{aligned} A \text{ (downwards)}: &\quad 2mg - T = 2ma_1, \\ B \text{ (upwards)}: &\quad T - (1+\lambda)mg = (1+\lambda)ma_1. \end{aligned}$$

Adding eliminates T:

$$(1-\lambda)mg = (3+\lambda)ma_1 \quad \Longrightarrow \quad a_1 = \frac{(1-\lambda)g}{3+\lambda},$$

which is positive because $\lambda < 1$, so A really does descend. Starting from rest and travelling a distance l with constant acceleration, $l = \frac{1}{2}a_1 t_1^2$, so

$$t_1 = \sqrt{\frac{2l}{a_1}} = \sqrt{\frac{2l}{g}}\sqrt{\frac{3+\lambda}{1-\lambda}}.$$

Ask yourself In stage 2 bucket A has no mass at all. What is the tension in the chain?

Stage 2: B descends. Now A is empty, so its mass is zero, and B has mass m. For the massless bucket A, Newton's law reads $T - 0 \cdot g = 0 \cdot a$, so $T = 0$: a light chain attached to nothing with mass can't pull on anything. So B falls freely with acceleration g from rest through a distance l, and $l = \frac{1}{2}gt_2^2$ gives

$$t_2 = \sqrt{\frac{2l}{g}}.$$

(The same answer comes from the Atwood formula with masses 0 and m: $a = \frac{m-0}{m+0}g = g$.) At the end of this stage A is back at the top for the first time.

Total. The time from releasing A at the top until it first returns to the top is

$$\boxed{t_1 + t_2 = \sqrt{\frac{2l}{g}}\left(1 + \sqrt{\frac{3+\lambda}{1-\lambda}}\right) = \sqrt{\frac{2l}{g}}\,\frac{(1-\lambda)^{1/2} + (3+\lambda)^{1/2}}{(1-\lambda)^{1/2}}}$$

Maximising the ore raised

Write $\tau(\lambda) = t_1 + t_2$ for the cycle time. Each cycle raises ore of mass λm, and the process repeats identically. In a long time S there are $\lfloor S/\tau \rfloor$ complete cycles, so the ore raised is between $\lambda m\,(S/\tau - 1)$ and $\lambda m\, S/\tau$. When S is very large compared with τ, the lost part-cycle is negligible, and the ore raised is essentially

$$\frac{\lambda m S}{\tau(\lambda)} = mS\sqrt{\frac{g}{2l}}\,\frac{\lambda(1-\lambda)^{1/2}}{(1-\lambda)^{1/2} + (3+\lambda)^{1/2}} = mS\sqrt{\frac{g}{2l}}\,\mathrm{f}(\lambda).$$

The factor $mS\sqrt{g/2l}$ doesn't depend on λ, so we must choose $\lambda \in (0,1)$ to make $\mathrm{f}(\lambda)$ as large as possible.

Ask yourself Why must the best λ be a stationary point, and not an end point?

The function f is differentiable on $(0,1)$, since $1-\lambda>0$ and $3+\lambda>0$ there and the denominator is positive. It is positive on $(0,1)$, and it tends to 0 at both ends: as $\lambda\to 0$ the factor λ vanishes (no ore is carried), and as $\lambda\to 1$ the factor $(1-\lambda)^{1/2}$ vanishes (the buckets nearly balance, so stage 1 takes a very long time). So the largest value of f is reached at an interior point, and a maximum of a differentiable function at an interior point has zero derivative. So the best λ satisfies $\mathrm{f}'(\lambda)=0$, as required.

Checks

- Limiting case: as $\lambda\to 1$ the buckets in stage 1 have masses $2m$ and $2m$, so $a_1\to 0$ and $t_1\to\infty$, which matches the formula.
- As $\lambda\to 0$ we get $a_1=g/3$, the usual Atwood value for masses $2m$ and m.
- Numerically, stepping the motion with $g=9.8$ for random λ and l agreed with the formula: for example $\lambda=0.504$, $l=26.9$ gives a cycle of 8.568 s. Maximising f numerically gives $\lambda\approx 0.704$, and counting whole cycles in a long time gives the same best λ.

Common slips

- Forgetting the mass m of bucket B itself, and using λm instead of $(1+\lambda)m$ in stage 1.
- Treating stage 2 as an Atwood machine with some tension: with A massless, B falls freely.
- Maximising ore per cycle, λm, instead of ore per unit time. Bigger loads mean slower cycles.

Beyond STEP

Optimising a rate (output per cycle divided by time per cycle) is the idea behind the *renewal–reward theorem* in probability, where cycle lengths are random and the long-run reward rate is still the mean reward per cycle divided by the mean cycle length. The mechanics here is the constant-acceleration motion that Kleppner and Kolenkow, *An Introduction to Mechanics*, develops into a full treatment of systems of particles.

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